IROS 2026
PITTSBURGH

2nd Workshop on Geometric-Aware Representations: Structured priors, metrics, constraints, and groundings Manipulation

1. Organizers

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The organizing team has strong experience in organizing international workshops, summits, and technical events across robotics, manipulation, learning, and geometric methods. Importantly, this workshop is the second edition of a topic that has already shown clear community demand: the first workshop on geometric representations, held at ICRA 2023, attracted over 120 participants and generated extensive discussion and interaction. Building on that success, this new edition shifts the focus from the representation formalisms themselves toward their applications across the manipulation stack, with particular emphasis on inductive bias, grounding, and metric design. For learning, geometry shapes what models can represent, what solutions they prefer, and how reliably they compose with downstream planners and controllers. For planning and control, geometry provides principled ways to encode feasibility, interaction intent, and contact consistency through structured costs and constraints. In addition, Dr. Luis Figueredo has also co-organized the 3rd UK Manipulation Workshop in 2019, then the largest UK meeting in robotic manipulation with participation from over 32 institutions, and is also the founder of the Geriatronics Summit, now approaching its fifth edition, with the latest event attracting more than 220 participants. Together with Prof. Bruno Adorno, he also co-organized the IEEE ICAR 2019 workshop on dual quaternion algebra in robotics. Riddhiman Laha and Tobias Löw co-organized the upcoming ICRA 2026 workshop on geometry in data-driven robotics, while Laha additionally organized an RSS 2025 workshop on contact-rich manipulation. Prof. Nilanjan Chakraborty contributes experience from prior ICRA workshops and a two-day ASME IDETC summer school, Prof. Sylvain Calinon has co-organized around 20 workshops over the last decade, and Prof. Seth Hutchinson adds long-standing leadership across major robotics venues, including xxxxxxxxxxxxxx. Altogether, the team combines a strong technical fit with a proven ability to curate interdisciplinary, discussion-oriented events that can genuinely catalyze new collaborations and research directions.

2. Abstract and Website

Website: xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx

Recent advances in robotic manipulation have enabled increasingly tight integration between perception, planning, learning, and control. Despite this progress, the role of geometric information, such as representations, structures, metrics, priors, and constraints, remains fragmented across these subareas. While geometry is fundamental to reasoning about objects, modeling, contacts, and physical interactions, it is often treated as an implementation detail or modeling convenience rather than as a unifying framework for manipulation.

Prior workshops have typically focused either on geometric representations themselves (as in our previous workshop at ICRA 23) or on specific downstream applications, such as contact-rich planning or grasping. As a result, there is still no shared language for incorporating geometric structure throughout the full manipulation pipeline. This workshop aims to bridge that gap by examining how geometric-aware information can serve as a common foundation connecting perception, planning, learning, and control.

We propose to bring together researchers working on structured priors, geometric metrics, constraint-based reasoning, and physically grounded representations for manipulation. The goal is to highlight how geometric structure can guide learning, improve generalization, and enable more robust reasoning about interactions with the physical world. Through invited talks, contributed presentations, and focused discussions, the workshop will explore emerging directions for integrating geometry as a first-class component of manipulation systems rather than merely a modeling convenience.

3. Content of the workshop & expected impact

This workshop aims to establish geometric methods and representations not as a narrow modeling formalism, but as a practical language for modern robotics. Building on the strong interest demonstrated by the first edition at ICRA 2023, which attracted more than 120 participants, this second edition shifts the focus from representation formalisms alone to the role of geometric structure across the full manipulation stack: sensing and state estimation, learning and latent representations, planning and optimization, and control under contact, physical and geometric constraints. Its central objective is to clarify when geometry should enter as grounding, as inductive bias, as a task-relevant metric, and as an explicit constraint, while creating a shared vocabulary across communities that often study related problems using different assumptions and terminology. This framing is timely because robotic manipulation is increasingly built from tightly coupled perception, learning, planning, and feedback components, yet the role of geometry remains fragmented across these layers.

Describe the objectives of the workshop

This is the second edition of our workshop on geometric aware representations. This workshop seeks to foster in-depth discussions, explore new ideas, and create new connections in the field of intelligent robotic manipulation planning and control. The primary objective is to bring together experts from different fields of robotics, AI, and applied mathematics to explore the latest advancements in their area, together with novel solutions for exploiting geometric information in robotics and AI-driven learning methods, and control methodologies, as well as their application and impact on single and multi-arm manipulation.

The workshop aims to catalyze dynamic discussions and tackle the pressing challenges hindering the entry of geometric methods as a language in software deployment of single and multi-arm systems in practical settings, while also contributing to the theoretical development of the subfield. Through guided interdisciplinary discussions, the event will accelerate tangible progress in geometry-as-a-language solutions for industry, service robotics, and healthcare, where geometric structure can guide learning, improve generalization, and enable more robust reasoning about interactions with the physical world.

The workshop will structure discussions around key challenges such as defining a comprehensive taxonomy of geometric concepts for modelling the configuration and the operational space, investigating geometric priors for fast online learning, and exploring efficient representations for physically grounded robots. This workshop will facilitate forward-looking discussions that will shape the future of geometric learning, planning, and control and go beyond geometry as a modeling convenience. The workshop is expected to attract a diverse group of researchers, experts, and practitioners, fostering a holistic and interdisciplinary perspective on

these challenges. Attendees will have the opportunity to share their latest findings, innovative concepts, and real-world applications, ensuring that discussions represent the most cutting-edge advancements in the field. Within this framework, the workshop will play a pivotal role in inspiring

junior researchers, strengthening knowledge exchange between established experts, and laying the foundation for future collaborative efforts in geometric information for robots.

To ensure the highest quality content, the workshop will feature a carefully curated selection of invited speakers, panel discussions, and interactive sessions designed to promote deep engagement. Leading researchers will provide state-of-the-art insights, highlighting challenges and novel frameworks, along with experimental results focused on real-world deployment. In contrast to standard workshop formats, a key component of this event will be the Interdisciplinary Impact Clinics (IIC), where early-career researchers and students will receive tailored feedback from senior experts to refine their approaches and maximize real-world impact through interdisciplinary collaboration. These clinics will be made possible through a careful selection of invited speakers, who will actively engage in interactive mentoring and discussions. Additionally, video demonstrations and poster sessions will allow participants to directly interact with state-of-the-art multi-arm robotic technologies, gaining insights into their capabilities, limitations, and future directions. Structured panel discussions and roundtables will further encourage open debates on challenges, solutions, deployment, and research collaboration opportunities in geometric robotics.

Expanding the scope of IROS 2026 content, this workshop will integrate expertise from robotics, AI, and control to examine geometry-aware representations from multiple perspectives. The invited speakers represent a diverse group of experts from different fields, ensuring a balanced discussion, fostering interdisciplinarity and bringing different views for theoretical innovations and real-world deployment. The workshop strongly promotes participation from underrepresented groups and early-career researchers, fostering inclusivity and broadening engagement across the robotics community.

Aligned with the …..

Explain how you will ensure the highest quality content:

- Does the content reflect the state of the art in a subfield of robotics and automation?
- How will you engage the attendees and exchange ideas?
- How will the proposed event expand the originality and diversity of workshop content at IROS 2026?

4. Intended audience & expected attendance

The call for contributions and participation will target a broad range of communities working on robotic manipulation, robot learning, planning and control, geometric robotics, state estimation, physically grounded perception, and embodied AI. Overall, we expect this workshop to attract researchers interested in both the foundational and applied use of geometric structure in robotic systems, including structured priors, geometric metrics, constraints, groundings, and manifold-aware representations. By bringing together perspectives from

sensing, learning, planning, and control, the workshop aims to provide a holistic view of how geometric-aware information can improve generalization, composability, robustness, and physical consistency in manipulation systems, while also clarifying the advantages and limitations of different structured approaches in practice.

Given the strongly cross-disciplinary backbone of the proposal, we will seek to attract both academic and industrial partners connected to several relevant IEEE RAS TCs and adjacent research communities. In particular, the workshop is expected to be of interest to researchers affiliated with the IEEE RAS TCs on Robot Learning, Whole-Body Control, Human-Robot Interaction & Coordination, Robot Control, and Model-Based Optimization for Robotics, as well as to communities working more broadly on robot manipulation, geometric control, computer vision for robotics, tactile sensing, motion planning, state estimation, dexterous manipulation, and mobile manipulation. Finally, we will also reach out to the RAS Student Activities Committee to encourage student participation and strengthen interaction between early-career researchers and established experts. This follows the same inclusive and cross-community philosophy successfully used in previous workshops organized by members of the team.

Based on the success of previous workshops and summer schools organized by the team at ICRA, IROS, and related IEEE venues, and particularly the successful event of the first version of the workshop, we therefore expect an attendance of approximately 70–120 participants. This will likely include faculty, industry researchers, postdoctoral fellows, and graduate students, providing a sufficiently broad yet interactive environment for discussion. Similar attendance ranges were achieved in prior events organized by members of the team, particularly in workshops that connected multiple subcommunities around a common methodological theme.

5. Invited Speakers

Given our organizing team and selected speakers, we believe that this workshop is well positioned to bring together researchers working on complementary uses of geometric structure across manipulation, from state and representation design to learning, planning, and control. To ensure a clear and balanced organization of the content, the invited speakers are grouped according to the main thematic blocks of the workshop. Each block is intended to combine complementary perspectives and will be followed by a focused panel discussion.

(i) Geometric priors and groundings for sensing

→ **Prof. Joan Lasenby** (**confirmed**)**,** Professor, University of Cambridge, UK

Website: https://www.eng.cam.ac.uk/profiles/jl221

→ **Prof. Georgia Chalvatzaki (confirmed),** Professor, Technical University of Darmstadt, Germany

Website: https://pearl-lab.com/people/georgia-chalvatzaki/

(ii) Structure-preserving learning and manifold-valued representations

→ **Arne Sachtler (confirmed),** Doctoral Candidate, TU Munich – German Aerospace Center (DLR).

Website: https://www.ce.cit.tum.de/rm/people/arne-sachtler/

→ **Prof. Constantinos Chamzas (confirmed)** , Assistant Professor, Worcester Polytechnic Institute, USA. Website: https://www.wpi.edu/people/faculty/cchamzas

(iii) Geometry-aware planning, optimization, and control

→ **Prof. Robert Mahony** (**confirmed**)**,** Professor**,** Australian National University, Australia

Website: https://eng.anu.edu.au/people/robert-mahony

→ **Dr. Johannes Lachner (confirmed),** Postdoctoral Researcher, Massachusetts Inst. of Technology, USA. Website: https://jlachner.github.io/

(iv) Geometric world models, affordances, and generative representations for manipulation

→ Prof. **Jen Jen Chung** (**Invited**) , Professor, University of Queensland, Australia

Website: https://eecs.uq.edu.au/profile/8084/jen-jen-chung

→ **Prof. Francesco Bullo (invited)** , Professor, University of California, Santa Barbara, USA

Website: https://fbullo.github.io/

This four-block organization gives the workshop a clear internal logic while preserving the breadth needed to engage multiple communities. It also supports one of the workshop's main goals: to compare how geometric structure is used differently across the manipulation stack — as grounding, inductive bias, representation, mechanics, metric, and constraint — and to foster discussion on how these viewpoints can be better connected in future robotic systems.

6. Structure of the workshop

This workshop will feature five complementary types of sessions, carefully designed to foster in-depth technical discussion, exchange across communities, and active interaction between senior experts, early-career researchers, and practitioners. The central objective is not only to present recent advances in geometric-aware representations and the usage of geometric methods for manipulation, but also to create a space where researchers can openly discuss how and where geometry should enter the manipulation stack—whether as structured priors, task-relevant metrics, symmetry-preserving representations, grounding mechanisms, or explicit constraints. The five complementary types of sessions are further detailed below.

(i) Keynote Presentations: Foundations in Geometric-Aware Manipulation

A select group of invited leading academics and expert researchers will present their perspectives, research, ideas, and open problems concerning the role of geometric-aware representations in robotic manipulation. The invited talks will be organized around four major themes, each reflecting a critical perspective on how geometry contributes to modern manipulation systems. **(i) Geometric priors and groundings for sensing:** Exploring object-, scene-, and contact-aware representations for manipulation, including structured state estimation, uncertainty on SE(3) and related manifolds, tactile and physically grounded perception, and object-centric or scene-centric representations that support downstream action. **(ii) Structure-preserving learning and manifold-valued representations:** Focusing on the use of geometric structure as inductive bias in learning systems, including equivariant models, symmetry-aware perception-to-action mappings, manifold-valued skills, manipulability and compliance representations, and structured latent spaces for interaction and adaptation. **(iii) Geometry-aware planning, optimization, and control:** Covering structured costs, task-relevant metrics, contact and kinodynamic constraints, geometric mechanics, and optimization/control frameworks that make geometry explicit in the computation of feasible, robust, and physically meaningful manipulation behavior. **(iv) Geometric world models, affordances, and generative representations for manipulation.** Exploring object-centric and scene-centric world models, affordance-grounded task representations, geometric latent spaces, and generative models on SE(3) and related manifolds for pose, trajectory, and interaction hypothesis generation. This block highlights how geometry can support prediction, reasoning, and transfer across objects, environments, and embodiments.

To encourage exchange of ideas and broader discussion, each invited speaker will be encouraged to present beyond a single paper or project, providing a wider perspective on lessons learned, foundational assumptions, trade-offs, and future directions. This is particularly important for the present workshop, as one of its main goals is to connect communities that often study similar structures from different perspectives but do not always use the same language to describe it. The workshop is expected to feature 8 invited talks, with most speakers already confirmed. Rather than treating them as isolated presentations, the talks will be grouped into thematic blocks so that each set of speakers can collectively frame a broader discussion.

(ii) Thematic Panel Discussions

Following each keynote block, curated short panel discussions will provide a platform for focused exchange between speakers and participants. These discussions will not simply serve as Q&A sessions, but as

opportunities to compare viewpoints across methods, assumptions, and application contexts. Each panel will revolve around a central question relevant to the workshop. These include some of the following.
How should geometric structure be represented in sensing and perception? What is gained by grounding representations in object pose, contact geometry, and physically meaningful state variables? Where do such grounded models remain brittle under uncertainty, occlusion, or partial observability? When should geometry enter learning as inductive bias? How much structure should be embedded directly into the representation, architecture, or loss? When do equivariance, manifold constraints, or geometric priors improve generalization, and when do they become overly restrictive? How do costs, metrics, and constraints shape planning and control? How can geometry-aware objectives and feasibility constraints be integrated into optimization and feedback control without sacrificing flexibility, robustness, or tractability?

A key aspect of these panels will be identifying conceptual and methodological barriers between geometric robotics, robot learning, perception, and control communities, and using those discussions to establish a more unified perspective on structure-aware manipulation.

(iii) Interactive Poster Sessions and Poster Teaser Presentations

To engage the wider research community, the workshop will include an **open call for 2–4 page extended abstracts**, inviting participants to present work related to the workshop themes. Accepted contributions will be given the opportunity to present their work both in a short teaser format and in an interactive poster session. Relevant topics will include, but are not limited to:

- Geometric world models and object-centric scene representations for manipulation;
- SE(3)-aware perception, tracking, and state estimation for interaction;
- Geometry-aware latent spaces and structured representation learning;
- Task and motion planning with geometric affordances and relational structure;
- Geometric mechanics and motion manifolds for robot modelling and control;
- Signed distance fields, implicit geometry, and contact-aware scene encoding;
- Tactile and visuo-tactile grounding through geometric contact representations;
- Geometric reasoning for dexterous and multi-finger manipulation;
- Whole-body and mobile manipulation with structure-aware representations;
- Geometric representations for compliance, stiffness, and impedance shaping;
- Hybrid learned/model-based architectures for manipulation under constraints;
- Cross-embodiment and cross-task transfer through structured geometric representation;

Presenters of accepted abstracts will give a short spotlight teaser presentation to introduce their work to the full audience and invite participants for follow-up discussion during the poster sessions. The main objective is to provide a strong platform for **young researchers and early-career participants** to present ongoing research, sharpen their ideas, and receive direct feedback from leading researchers in the field. As in previous workshops organized by the team, special attention will be given to encouraging participation from **women and early-career researchers**, helping broaden representation in a technically demanding and often methodologically fragmented area.

(iv) Workshop Clinics: From Method Design to System Integration

A distinctive feature of the workshop will be a set of short **Geometric Representation Clinics**, directly linked to the accepted contributions and poster presenters. These clinics are especially aimed at early-career researchers and contributors who are developing new representations, models, or methodologies, but who would benefit from targeted feedback on how their ideas position themselves within the broader manipulation stack. In these sessions, senior speakers and organizers will provide tailored feedback to questions from the early-career contributions. These will be obtained from a previous questionnaire-form. In addition, senior members and invited speakers will be asked to comment on each early career contribution, answering

questions specific to their works with focus on: (i) Is the geometric structure being used primarily as an inductive bias, as a grounding mechanism, or as an explicit constraint? (ii) What assumptions are embedded in the method, and how strongly do they influence generalization? (iii) Does the representation compose naturally with downstream planning or control? (iv) What kinds of experiments, ablations, or deployment examples would strengthen the contribution?

This format is intended to create a more direct mentoring and exchange opportunity than standard Q&A, while also helping participants better articulate the role of geometric structures, as priors and constraints, and usage of valid and relevant metrics in their own methods.

(v) Round-Table Discussion: Inductive Bias vs Grounding in Manipulation

To foster deeper dialogue and idea exchange, the workshop will conclude with a round-table discussion involving all speakers, organizers, and participants. This final session will focus on one of the workshop's most important conceptual questions: **How should geometric structure be used in manipulation systems: as inductive bias for learning, or as grounding and explicit constraints for physical reasoning and control?**

This discussion will explicitly compare the two viewpoints and explore their advantages, limitations, and appropriate domains of use. On one side, geometry can be introduced as **inductive bias** in learning systems through equivariance, manifold-aware features, structured latent spaces, or geometry-consistent losses. This can improve generalization, reduce data requirements, and guide learning toward more meaningful or composable solutions. However, such priors can also be overly restrictive, may break under symmetry-breaking conditions, and can hide mismatches between abstract structure and physical deployment. On the other side, geometry can be used as **grounding and constraint** through contact models, SE(3)-consistent state representations, feasible sets, task-relevant metrics, signed-distance constraints, or physically meaningful optimization objectives. These approaches often offer stronger physical interpretability and better guarantees on feasibility and consistency, especially in contact-rich tasks. Yet they may also be less flexible, more model-dependent, and harder to deploy under uncertainty, noisy sensing, or incomplete knowledge.

The round-table will therefore focus on (i) when each viewpoint is most appropriate, (ii) how they can be combined rather than opposed, (iii) what kinds of success stories already demonstrate this hybridization, (iii) and what new opportunities arise when structured learning and geometric control are designed together.

We expect this discussion to be particularly valuable because it moves beyond simply listing methods and instead addresses the deeper design choices that currently shape the field. To support this, pre-workshop questions will be collected from participants through an online form and revisited during the final session. The round-table will therefore serve not only as a summary of the workshop, but also as a forward-looking discussion on the future of geometry-aware manipulation. This structured format aims to catalyze collaboration across robotics subfields, strengthen the role of geometric methods in real manipulation systems, and define a broader research agenda for integrating structured priors, metrics, constraints, and grounding throughout the manipulation pipeline.

7. Tentative Schedule

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09h00	— 10 min	Welcome by Organizers
09h10	— 30 min	**Speaker 1**:
09h40	— 30 min	**Speaker 2**:
10h10	— 30 min	Poster/Video teasers
10h40	— 30 min	Coffee Break
11h10	— 30 min	**Speaker 3**:
11h40	— 10 min	**Panel Discussions 1:** Interdisciplinary Perspectives on Modeling and Design
11h50	— 30 min	**Speaker 4**: Johannes Lachner - Energy Budgets for Safe Interactions
12h20	— 30 min	**Speaker 5**: Arne Sachtler - Connection between Riemannian manifolds and mechanics
12h50	— 70 min	Lunch Break/Networking
14h00	— 30 min	**Speaker 6:**
14h30	— 10 min	**Panel Discussions 2:**
14h40	— 30 min	**Speaker 7**:
15h10	— 30 min	**Speaker 8**:
15h40	— 30 min	Poster-Session — Coffee Break/Networking
16h50	— 20 min	
17h10	— 20 min	Round-Table Discussion: Future Directions in Safe Human-Robot Interaction

8. Plan to solicit participation

To ensure strong participation from researchers across the different communities relevant to this workshop, we have developed a comprehensive outreach plan that extends beyond traditional mailing-list announcements. Our goal is to attract a diverse and engaged audience spanning robotic manipulation, robot learning, geometric robotics, planning and control, perception, embodied AI, and related areas. Our strategy is as follows.

(a) Multi-Channel Outreach: We recognize the importance of using multiple communication channels to reach potential participants. In addition to direct email announcements and distribution through mailing lists such as robotics-worldwide, we will actively promote the workshop through social media platforms such as LinkedIn, X, and other professional online channels. Regular posts, speaker announcements, and deadline reminders will be used to maintain visibility over time. As this workshop brings together several research communities that do not always interact directly, we will also target relevant forums, academic communities, and online groups in robotics, control, machine learning, and computer vision.

(b) Actively publicize the workshop through social media and professional online platforms, e.g., LinkedIn, X, and research-group websites. Speaker announcements, workshop themes, and deadlines for contributed abstracts will be shared progressively to maintain visibility over time and to attract both senior researchers and early-career participants. The workshop website will serve as the main reference point for all updates, submission information, and the program.

(c) Third, we will reach out to relevant **IEEE RAS Technical Committees** and adjacent research communities whose members are likely to be interested in the workshop themes. In particular, we expect strong relevance to communities in robot learning, manipulation, robot control, whole-body control, human-robot interaction, and

model-based optimization, as well as to works on geometric methods, state estimation, and physically grounded perception.

(d) Direct Engagement with Universities and Research Institutes: We will directly engage with universities, research institutes, and labs known for their work in robotic manipulation, geometric methods, robot learning, motion planning, state estimation, embodied AI, and physical human-robot interaction. This will involve contacting professors, researchers, postdoctoral fellows, and group leaders who may have students or collaborators interested in participating. Through the organizers' and invited speakers' networks, we expect this direct outreach to be particularly effective in bringing together participants from complementary methodological backgrounds.

(e) Industry and Applied Research Engagement: To attract researchers and practitioners with real-world expertise, we will also reach out to industrial research labs, robotics companies, and applied research centers working on manipulation, autonomy, robot learning, and physically grounded robotic systems. Participation from industry is especially valuable for this workshop, as many of the core questions concern how structured priors, constraints, and grounding can improve robustness and composability in deployed systems. This interaction can also help connect foundational research to practical deployment needs.

(f) Conference and Community Announcements: We will leverage academic conferences, workshops, and community events leading up to IROS 2026 to promote the workshop. This includes disseminating information through organizers' and speakers' presentations, existing collaborations, and related research meetings in robotics and AI. Where possible, we will also use community-facing opportunities, such as lab seminars or partner events, to raise awareness of the workshop and encourage contributions.

(g) Personalized Invitations: We will send personalized invitations to researchers who have previously shown interest in closely related topics, have attended similar workshops, or are active in areas directly connected to the workshop themes. These targeted invitations will be particularly important for drawing together experts from communities such as geometric robotics, manipulation learning, tactile/contact-rich interaction, and optimization-based control, who may not always respond equally to broad announcements.

(h) Engagement with Online Communities: Active participation in online robotics and AI communities, forums, and discussion spaces will also be part of our strategy. We will share workshop information, engage with discussions relevant to the workshop themes, and encourage community members to circulate the call for participation. This approach is especially useful for reaching early-career researchers and interdisciplinary groups who may encounter the workshop through topic-based communities rather than through conference mailing lists alone.

(i) Early Communication: We will circulate "save the date" announcements and early calls for participation well in advance, giving potential attendees enough time to plan their travel, contributions, and conference schedule. Early communication is particularly important for a workshop of this kind, where participation may come from researchers across multiple subfields who need time to assess the workshop's relevance to their own work.

(j) Participant Benefits: We will highlight the distinctive benefits of participating in the workshop, including the opportunity to engage with leading researchers across sensing, learning, planning, and control; to discuss lessons learned on structured representations and geometric priors; and to exchange ideas on how geometry can improve robustness, generalization, and composability in real robotic systems. We will also emphasize the visibility offered by the teaser talks and poster sessions, especially for students and early-career researchers, as well as the workshop's focus on discussion, interaction, and community building rather than only formal presentations.

9. Dissemination

Our dissemination plan for the contributions to the workshop aims to maximize the reach, accessibility, and long-term usefulness of the content, ensuring that it benefits a broad audience, including researchers from communities that are not always equally represented across robotics, learning, and geometric methods. We will employ a multifaceted approach to disseminate materials, encourage continued discussion, and create an

accessible online archive centered around the workshop theme of **geometric-aware representations for manipulation**.

Online Repository: We will maintain a dedicated workshop website that will serve as the central hub for the event. This platform will host practical information about the workshop, the list of invited speakers, the accepted contributions, and post-workshop materials. As in our previous workshops, the website will remain available after the event to ensure long-term access to the content.

Media Archive: Where permission is granted by the speakers and authors, presentation slides, posters, and video recordings will be made available on the workshop website. Participants will be able to access these resources before, during, and after the event, allowing the workshop to remain useful as a reference point for both attendees and researchers who could not participate in person. We also aim to include a brief bibliography of representative references related to the workshop themes, to facilitate cross-pollination between the different communities involved.

Social Media Engagement: We will leverage professional and social media platforms such as LinkedIn, X, and other academic channels to disseminate workshop updates, speaker, highlights, and key takeaways. Participants will be encouraged to share insights and discussion points online, helping broaden the digital footprint of the workshop and attract a wider audience interested in manipulation, robot learning, planning, perception, and geometric methods.

Collaboration with Partner Organizations and Research Networks: The workshop will be disseminated through the academic networks of the organizers and speakers, their home institutions, and their collaborators across robotics, control, machine learning, and manipulation. We will also seek support from relevant IEEE RAS Technical Committees and research communities aligned with robot learning, manipulation, whole-body control, human-robot interaction, and geometric robotics. This will allow the workshop materials to reach both established communities and adjacent audiences who may not normally attend methodology-focused workshops of this kind.

Open Access Policy: All workshop materials that can be shared with consent, including talk abstracts, accepted extended abstracts, posters, and slides, will be made openly accessible through the workshop website. This open-access approach is important for promoting knowledge exchange across communities that often remain separated, and for ensuring that the outcomes of the workshop can continue to inform future research after the event.

Post-Workshop Report and Follow-up: After the workshop, we will compile a concise report summarizing the key insights, recurring discussion points, and emerging trends that arose during the invited talks, poster interactions, and the final round-table discussion. In particular, this report will synthesize the different perspectives discussed during the workshop on the use of geometric structure as inductive bias, grounding, metrics, and constraints across the manipulation stack. The report will be made available on the workshop website and circulated through relevant mailing lists and research communities. We also intend to offer participants the possibility of registering for follow-up communication regarding future workshop editions, related initiatives, and possible post-workshop dissemination opportunities.

Engagement with Underrepresented Groups: To reach a broader and more diverse audience, we will actively promote the workshop through academic networks, mailing lists, forums, and organizations that support diversity and inclusion in robotics and AI. We will also emphasize the participation of women and early-career researchers in both the invited and contributed components of the workshop, and, where feasible, we will support broader inclusion through targeted outreach and visibility efforts. This is particularly important for a workshop that aims to connect communities across technical and disciplinary boundaries.

By implementing these strategies, we aim to create an inclusive and accessible online archive that promotes active engagement, long-term knowledge sharing, and continued interaction beyond the day of the workshop itself. Our commitment to open dissemination and targeted outreach will help ensure that the workshop's contributions benefit a diverse global audience, including researchers working at the intersection of manipulation, control, learning, perception, and geometric robotics.

9.1 Contributed Papers

We will solicit **extended abstracts and position papers of 2 to 4 pages**, with an emphasis on technically grounded contributions that fit the workshop themes. We will also encourage the submission of videos, demos, or visual material accompanying the abstracts, especially when they help illustrate geometric structure, contact-rich interaction, or representation-driven manipulation behavior. All submissions will undergo peer review, with attention to relevance, technical quality, novelty, and their potential to stimulate discussion during the workshop. Accepted contributions will be presented through spotlight teaser talks and interactive poster sessions, giving contributors visibility and direct access to feedback from invited speakers and attendees.

10. Equipment

To facilitate our workshop, we will need approximately 15 poster boards to accommodate interactive presentations of accepted papers. Additionally, we will require standard audio-visual equipment, such as a projector or a TV panel equipped with microphones for the speakers. If possible, a room layout that facilitates discussion during breaks would be highly beneficial. No unusual power or demo requirements are currently anticipated, but we would welcome a flexible setup that can support short spotlight presentations and panel-style discussions comfortably.

Appendix – List of Accepted Invited Speakers – Acceptance Letters

Please find below a list of acceptance emails/letters from the invited speakers.

[IROS-WS-26] Workshop Proposal - Invitation for Keynote Speaker - Title: Geometric-Representations: Structured priors, metrics, constraints, and groundings for Manipulation Summarize

JL

Joan Lasenby<jl221@cam.ac.uk>

To: Luis Figueredo <figueredo@ieee.org>

Fri 3/13/2026 10:25 AM

Cc: Laha, Riddhiman; Yuhe Gong <yuhe.gong@nottingham.ac.uk>

You don't often get email from jl221@cam.ac.uk. Learn why this is important

Morning Luis

OK, yes, happy for you to put my name down. I've looked at the workshop description — I could talk about a number of things, is there any specific aspect you think would fit well and complement others?

Best

Joan

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Prof Joan Lasenby
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CC

Chamzas, Constantinos<cchamzas@wpi.edu>

To: Laha, Riddhiman

Cc: lfc.figueredo@gmail.com

Fri 3/13/2026 3:40 PM

Hi Riddhi,

Thanks for the invitation! Yes I am planning to attend IROS, and would love to participate as a speaker to the workshop!

Best,
Constantinos

Sent from Outlook for iOS

From: Laha, Riddhiman <r.laha@northeastern.edu>
Sent: Thursday, March 12, 2026 8:48 PM
To: Chamzas, Constantinos <cchamzas@wpi.edu>
Cc: lfc.figueredo@gmail.com <lfc.figueredo@gmail.com>
Subject: [EXT] [IROS WS 26] Workshop Speaker Invitation

> You don't often get email from r.laha@northeastern.edu. Learn why this is important

...

Hello Constantinos,

Hope all is well.

I am writing on behalf of the organising committee to kindly invite you to participate as a keynote speaker at the **2nd Workshop on Geometric-Representations: Structured priors, metrics, constraints, and groundings for Manipulation**, which we are organising for **IROS 2026** in Pittsburgh (Sept 27 – Oct 1).

****EXTERN** [IROS-WS-26] Workshop Proposal - Invitation for Keynote Speaker - Title: Geometric-Representations: Structured priors, metrics, constraints, and groundings for Manipulation** Summarize

GC

To: Luis Figueredo <figueredo@ieee.org>

Cc: Laha, Riddhiman; Yuhe Gong <yuhe.gong@nottingham.ac.uk>

Fri 3/13/2026 12:19 PM

> You don't often get email from georgia.chalvatzaki@tu-darmstadt.de. Learn why this is important

Hi Louis,

Thank you for the invitation I can tentatively accept.

Best,
Georgia
Georgia Chalvatzaki
Professor of Interactive Robot Perception & Learning (PEARL)
pearl-lab.com

Department of Computer Science
Technische Universität Darmstadt | hessian.AI

ERC Starting Grantee (SIREN) | AI Emmy Noether DFG Grantee | (iROSA) | ELLIS Scholar | Alfried-Krupp Stiftung Awardee | Chair, IEEE RAS Women in Engineering

On 13. Mar 2026, at 02:56, Luis Figueredo <figueredo@ieee.org> wrote:

Hi Georgia,

Hope all is well.

Georgia, I'm writing on behalf of the organising committee to kindly invite you to participate as a keynote speaker at the **2nd Workshop on Geometric-Representations: Structured priors, metrics, constraints, and groundings for Manipulation**, which we are organising for **IROS 2026** in Pittsburgh (Sept 27 – Oct 1).